

Project 1: Supervised Learning

Start Assignment

- Due Jul 11 by 11:59pm
- Points 100
- Submitting a file upload
- Available after Jun 29 at 11am

Project 1: Supervised Learning (Kaggle Project)

For Project 1, you will work on a **supervised learning problem** using a dataset from Kaggle. This project is designed to give you hands-on experience with the full machine learning workflow.

Project Requirements

1. Choose a Dataset

- Go to Kaggle and select a dataset tied to a **supervised learning task**:
 - Classification or Regression
- If you're unsure what to pick, you may use a benchmark dataset such as the Titanic - Machine Learning from Disaster.

2. Exploratory Data Analysis (EDA)

- Understand the dataset structure
- Visualize key patterns and relationships
- Identify missing values, outliers, and potential issues

3. Data Preparation

- Handle missing data
- Perform feature engineering if needed
- Encode categorical variables
- Normalize/scale features where appropriate
- Train/test split (or cross-validation)

4. Modeling

- Train at least one supervised learning model
- You are encouraged to try multiple models and compare performance

5. Evaluation

- Report appropriate performance metrics:
 - Classification: accuracy, precision, recall, F1-score, etc.
 - Regression: RMSE, MAE, R^2 , etc.
- Clearly interpret your results

6. Kaggle Participation

- Submit predictions to the Kaggle competition
- Include your Kaggle leaderboard score or results (if applicable)

7. Submission Requirements

- Upload your work to:
 - **Canvas** (notebook + any report)
 - **GitHub** (Share the link of your github on your notebook)
 - **Kaggle** (notebook and/or competition submission)
 - Make sure your code is clean, well-documented, and reproducible
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Notes

- You are free to choose any dataset **as long as it is supervised learning**
- If you're unsure about your dataset choice, reach out early
- Start early — good EDA and preprocessing take time